

**BURNET, TX, USA**WMO: **722542**Lat: **30.741N**Lon: **98.235W**Elev: **393**StdP: **96.70**Time Zone: **-6.00 (NAC)**Period: **97-19**WBAN: **03999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -3.2       | -1.4 | -12.3                         | 1.4 | 4.2  | -9.9 | 1.7 | 5.1  | 10.2                  | 12.2 | 9.0 | 13.2 | 2.9                   | 0    | 0.427 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 11.5                   | 37.3            | 22.0 | 36.2 | 22.2 | 35.0 | 22.3 | 24.5                | 30.8 | 24.1 | 30.2 | 23.8 | 29.8 | 3.7                  | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 22.9                            | 18.6 | 26.0 | 22.6 | 18.2 | 25.8 | 22.4 | 17.9 | 25.7 | 76.2          | 30.6 | 74.5 | 30.1 | 73.1 | 29.6 | 26.5              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| (a)               | (b)  | (c) | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| 8.5               | 7.7  | 7.0 | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
|                   |      |     | -6.6                       | 39.4 | 2.2                | 1.9 | -8.2                                               | 40.8 | -9.4       | 42.0 | -10.7      | 43.1 | -12.2      | 44.5 |  |  |
|                   |      |     | -8.0                       | 25.4 | 1.9                | 0.5 | -9.4                                               | 25.8 | -10.6      | 26.1 | -11.6      | 26.4 | -13.0      | 26.8 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 19.7   | 9.7   | 11.8  | 15.6  | 19.7  | 23.6  | 27.3  | 28.6  | 29.0  | 25.9  | 20.4  | 14.8  | 10.0 |
|                                                                                  | DBStd       | 8.01   | 5.21  | 5.58  | 4.75  | 3.89  | 3.18  | 2.03  | 1.71  | 1.90  | 2.89  | 4.37  | 5.04  | 5.06 |
|                                                                                  | HDD10.0     | 202    | 69    | 41    | 11    | 1     | 0     | 0     | 0     | 0     | 0     | 1     | 16    | 63   |
|                                                                                  | HDD18.3     | 1022   | 269   | 191   | 109   | 31    | 3     | 0     | 0     | 0     | 1     | 31    | 126   | 262  |
|                                                                                  | CDD10.0     | 3754   | 61    | 91    | 184   | 292   | 421   | 518   | 577   | 589   | 477   | 323   | 159   | 63   |
|                                                                                  | CDD18.3     | 1532   | 2     | 7     | 24    | 72    | 166   | 268   | 318   | 331   | 227   | 94    | 19    | 3    |
|                                                                                  | CDH23.3     | 16118  | 23    | 51    | 168   | 560   | 1427  | 2889  | 3782  | 4114  | 2263  | 723   | 104   | 15   |
|                                                                                  | CDH26.7     | 7631   | 2     | 8     | 27    | 158   | 541   | 1381  | 2003  | 2246  | 1036  | 216   | 13    | 1    |
| Wind                                                                             | WSAvg       | 3.2    | 3.1   | 3.4   | 3.7   | 3.9   | 3.7   | 3.5   | 3.1   | 2.8   | 2.6   | 2.9   | 3.0   | 3.1  |
| Precipitation                                                                    | PrecAvg     | 840    | 52    | 54    | 71    | 72    | 79    | 92    | 56    | 50    | 84    | 84    | 72    | 56   |
|                                                                                  | PrecMax     | 1203   | 134   | 160   | 219   | 183   | 177   | 287   | 234   | 155   | 217   | 290   | 272   | 247  |
|                                                                                  | PrecMin     | 543    | 1     | 3     | 15    | 7     | 16    | 4     | 0     | 0     | 1     | 0     | 1     | 2    |
|                                                                                  | PrecStd     | 202    | 40    | 42    | 44    | 45    | 47    | 69    | 55    | 43    | 53    | 72    | 70    | 54   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 26.2  | 27.7  | 29.1  | 32.8  | 35.0  | 37.6  | 38.0  | 38.9  | 37.8  | 32.4  | 28.3  | 25.3 |
|                                                                                  |             | MCWB   | 14.3  | 15.3  | 17.1  | 18.8  | 21.7  | 22.3  | 21.6  | 22.1  | 21.1  | 20.6  | 19.2  | 16.9 |
|                                                                                  | 2%          | DB     | 22.8  | 24.7  | 27.2  | 30.2  | 32.8  | 35.7  | 37.0  | 37.4  | 35.1  | 31.0  | 26.2  | 22.7 |
|                                                                                  |             | MCWB   | 14.4  | 14.8  | 17.0  | 19.1  | 21.8  | 22.5  | 21.9  | 22.2  | 21.6  | 20.8  | 18.0  | 15.8 |
|                                                                                  | 5%          | DB     | 20.9  | 22.5  | 25.3  | 28.2  | 31.4  | 34.0  | 35.9  | 36.3  | 33.4  | 29.0  | 24.0  | 20.9 |
|                                                                                  |             | MCWB   | 14.3  | 15.4  | 16.8  | 19.0  | 21.6  | 22.4  | 22.3  | 22.3  | 21.6  | 19.8  | 17.3  | 15.6 |
|                                                                                  | 10%         | DB     | 18.5  | 20.3  | 23.0  | 27.0  | 29.8  | 32.8  | 34.3  | 35.1  | 32.2  | 27.5  | 22.5  | 18.7 |
|                                                                                  |             | MCWB   | 13.7  | 15.0  | 16.0  | 18.5  | 21.0  | 22.5  | 22.4  | 22.5  | 21.5  | 19.4  | 17.2  | 14.5 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 19.0  | 19.7  | 20.5  | 22.9  | 24.6  | 25.3  | 24.9  | 24.8  | 24.4  | 23.9  | 21.9  | 19.7 |
|                                                                                  |             | MCDB   | 21.1  | 22.4  | 24.7  | 28.5  | 30.5  | 31.7  | 31.9  | 31.6  | 29.6  | 28.3  | 25.2  | 22.1 |
|                                                                                  | 2%          | WB     | 17.7  | 18.6  | 19.5  | 21.7  | 23.8  | 24.5  | 24.2  | 24.1  | 23.9  | 23.0  | 20.4  | 18.7 |
|                                                                                  |             | MCDB   | 19.9  | 21.3  | 23.1  | 26.4  | 29.1  | 30.7  | 30.7  | 30.7  | 29.0  | 26.6  | 23.1  | 21.0 |
|                                                                                  | 5%          | WB     | 16.2  | 17.7  | 18.8  | 20.9  | 23.1  | 24.0  | 23.9  | 23.9  | 23.5  | 22.3  | 19.4  | 17.2 |
|                                                                                  |             | MCDB   | 19.0  | 20.0  | 22.5  | 25.5  | 28.5  | 30.0  | 30.3  | 30.5  | 28.6  | 26.0  | 22.4  | 19.2 |
|                                                                                  | 10%         | WB     | 14.2  | 16.3  | 18.0  | 20.2  | 22.5  | 23.6  | 23.5  | 23.5  | 23.0  | 21.5  | 18.3  | 15.2 |
|                                                                                  |             | MCDB   | 17.3  | 19.4  | 21.4  | 24.4  | 27.5  | 29.4  | 29.8  | 30.2  | 28.1  | 25.2  | 21.3  | 18.2 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 11.8  | 11.5  | 11.6  | 11.5  | 10.7  | 10.6  | 11.2  | 11.5  | 11.0  | 11.2  | 11.0  | 11.4 |
|                                                                                  |             | MCDBR  | 15.2  | 14.0  | 13.6  | 13.1  | 12.4  | 12.2  | 13.0  | 13.2  | 12.8  | 12.0  | 12.4  | 13.7 |
|                                                                                  |             | MCWBR  | 8.7   | 7.7   | 6.7   | 5.8   | 4.3   | 2.9   | 2.7   | 2.8   | 3.5   | 4.9   | 6.5   | 8.6  |
|                                                                                  | 5% WB       | MCDBR  | 11.4  | 10.3  | 10.5  | 9.9   | 10.1  | 9.9   | 10.5  | 10.9  | 9.7   | 9.2   | 9.2   | 10.7 |
|                                                                                  | MCWBR       | 8.6    | 7.7   | 6.3   | 5.3   | 4.3   | 3.3   | 2.8   | 2.8   | 3.0   | 4.7   | 6.5   | 8.6   |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.310  | 0.318 | 0.345 | 0.385 | 0.418 | 0.411 | 0.423 | 0.414 | 0.388 | 0.362 | 0.336 | 0.313 |      |
|                                                                                  | taud        | 2.531  | 2.515 | 2.448 | 2.359 | 2.318 | 2.381 | 2.392 | 2.394 | 2.417 | 2.474 | 2.483 | 2.528 |      |
|                                                                                  | Ebn at Noon | 925    | 949   | 939   | 908   | 874   | 875   | 864   | 869   | 883   | 886   | 885   | 903   |      |
|                                                                                  | Edh at Noon | 86     | 95    | 109   | 124   | 130   | 122   | 120   | 119   | 112   | 99    | 90    | 82    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 3.09   | 3.65  | 4.65  | 5.57  | 5.86  | 6.73  | 6.75  | 6.33  | 5.22  | 4.25  | 3.24  | 2.74  |      |
|                                                                                  | RadStd      | 0.33   | 0.58  | 0.42  | 0.51  | 0.58  | 0.49  | 0.54  | 0.36  | 0.48  | 0.62  | 0.40  | 0.36  |      |

**Historical Trends**

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (40 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | +80     |

Nomenclature: See separate page